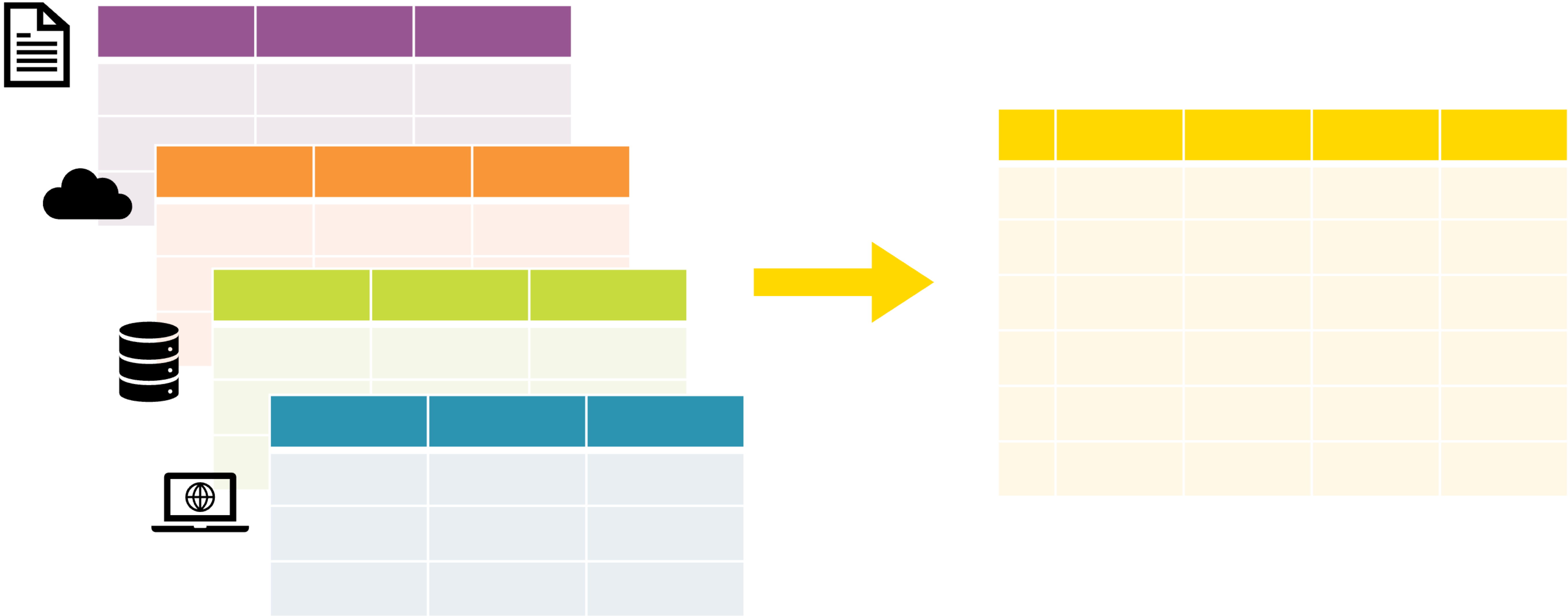
Mix and squeeze data

INTRODUCTION TO KNIME

Emilio Silvestri
Data Scientist

datacamp

Data merging



Value lookup

#	EmployeeID	Name	Department	Favorite food
1	EMP-43	Jenny	Sales	Pizza
2	EMP-29	Alex	Finance	Sushi
3	EMP-11	Taylor	R&D	Tacos
4	EMP-29	Alex	Finance	Sushi
5	EMP-66	Sam	R&D	Salad

Value lookup

#	EmployeeID	Name	Department	Favorite food
1	EMP-43	Jenny	Sales	Pizza
2	EMP-29	Alex	Finance	Sushi
3	EMP-11	Taylor	R&D	Tacos
4	EMP-29	Alex	Finance	Sushi
5	EMP-66	Sam	R&D	Salad

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup

#	EmployeeID	Name	Department	Favorite food
1	EMP-43	Jenny	Sales	Pizza
2	EMP-29	Alex	Finance	Sushi
3	EMP-11	Taylor	R&D	Tacos
4	EMP-29	Alex	Finance	Sushi
5	EMP-66	Sam	R&D	Salad

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	
2	EMP-29	Alex	Finance	Sushi	
3	EMP-11	Taylor	R&D	Tacos	
4	EMP-29	Alex	Finance	Sushi	
5	EMP-66	Sam	R&D	Salad	

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	
2	EMP-29	Alex	Finance	Sushi	
3	EMP-11	Taylor	R&D	Tacos	
4	EMP-29	Alex	Finance	Sushi	
5	EMP-66	Sam	R&D	Salad	

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	**28000**
2	EMP-29	Alex	Finance	Sushi	
3	EMP-11	Taylor	R&D	Tacos	
4	EMP-29	Alex	Finance	Sushi	
5	EMP-66	Sam	R&D	Salad	

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	**28000**
4	EMP-88	22000

Value lookup

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	28000
2	EMP-29	Alex	Finance	Sushi	22000
3	EMP-11	Taylor	R&D	Tacos	25000
4	EMP-29	Alex	Finance	Sushi	22000
5	EMP-66	Sam	R&D	Salad	

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Dictionary

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	28000
2	EMP-29	Alex	Finance	Sushi	22000
3	EMP-11	Taylor	R&D	Tacos	25000
4	EMP-29	Alex	Finance	Sushi	22000
5	EMP-66	Sam	R&D	Salad	

Lookup column

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Key column

Dictionary

Value lookup

#	EmployeeID	Name	Department	Favorite food	Salary
1	EMP-43	Jenny	Sales	Pizza	28000
2	EMP-29	Alex	Finance	Sushi	22000
3	EMP-11	Taylor	R&D	Tacos	25000
4	EMP-29	Alex	Finance	Sushi	22000
5	EMP-66	Sam	R&D	Salad	?

Lookup column

Key column

#	ID	Salary
1	EMP-11	25000
2	EMP-29	22000
3	EMP-43	28000
4	EMP-88	22000

Dictionary

Data aggregation

Data aggregation

Data aggregation

#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

aggregate

#	Sum of Salary
1	124000

Data aggregation

#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

Aggregation column

aggregate

#	Sum of Salary
1	124000

Data aggregation

#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

Aggregation column

Aggregation operation

aggregate

#	Sum of Salary
1	124000

Data aggregation

Category column

#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

aggregate

#	Department	Sum of Salary
1	Sales	28000
2	Finance	44000
3	R&D	52000

Data aggregation

#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

aggregate

#	Department	Number of employees
1	Sales	1
2	Finance	2
3	R&D	2

Data aggregation

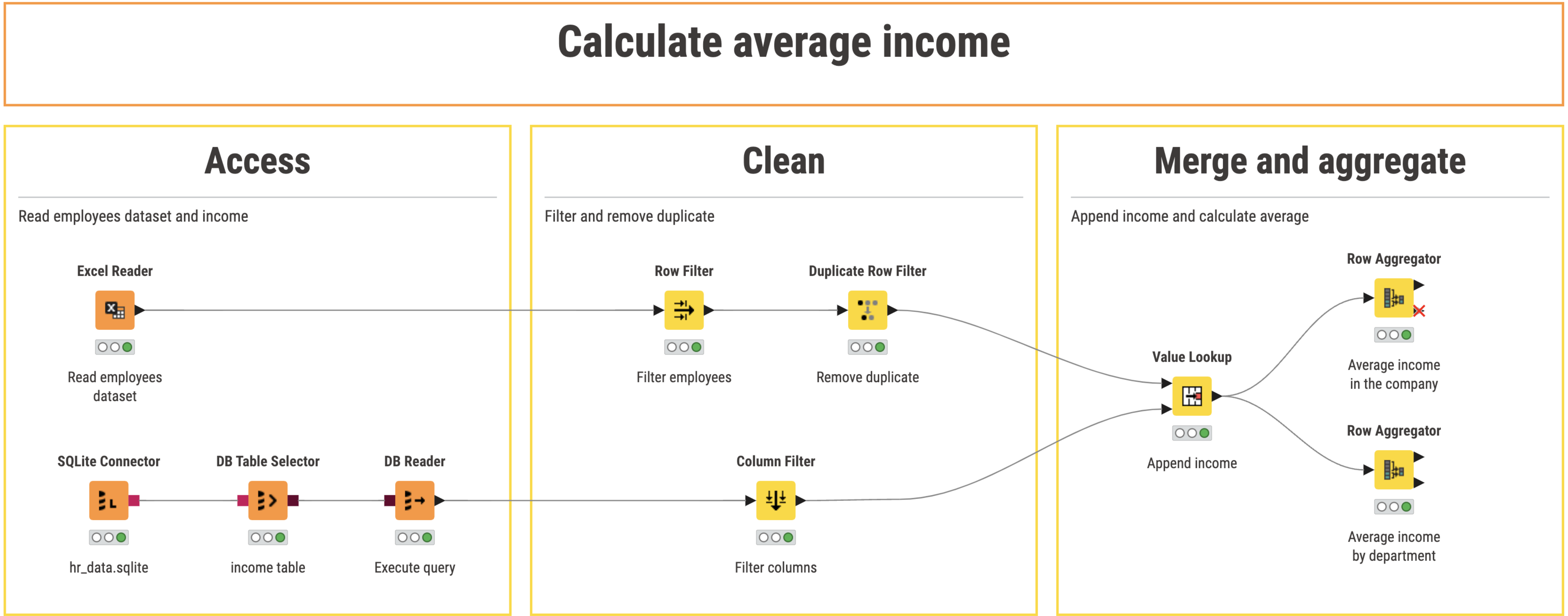
#	Name	Department	Salary
1	Jenny	Sales	28000
2	Alex	Finance	22000
3	Taylor	R&D	25000
4	Andrea	Finance	22000
5	Sam	R&D	27000

aggregate →

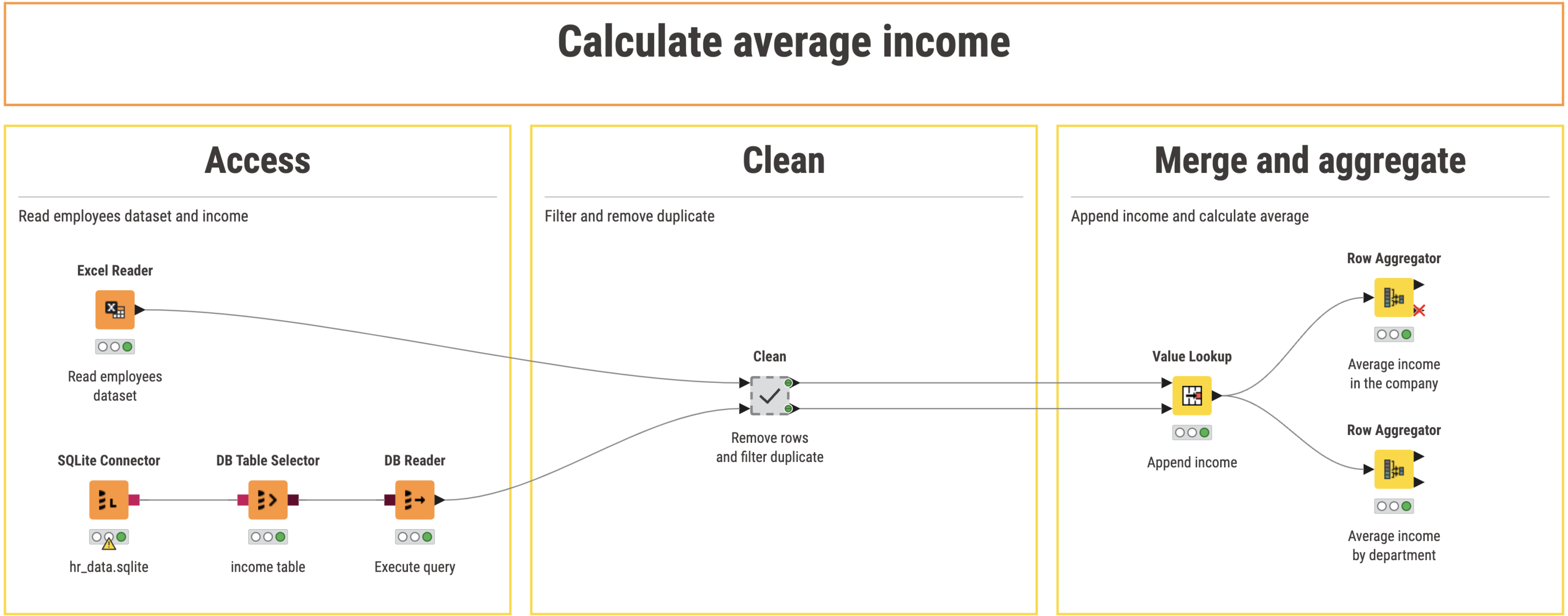
#	Department	Average Salary
1	Sales	28000
2	Finance	22000
3	R&D	26000

Documentation

Documentation



Metanodes



Calculate average income
Access
Read employees dataset and income
Excel Reader
Read employees dataset
SQLite Connector
hr_data.sqlite
DB Table Selector
income table
DB Reader
Execute query
Clean
Filter and remove duplicate
Clean
Remove rows and filter duplicate
Merge and aggregate
Append income and calculate average
Value Lookup
Append income
Row Aggregator
Average income in the company
Row Aggregator
Average income by department

Let's practice!

INTRODUCTION TO KNIME

Crunching data in KNIME Analytics Platform

INTRODUCTION TO KNIME

Emilio Silvestri
Data Scientist

Let's practice!

INTRODUCTION TO KNIME

Congratulations!

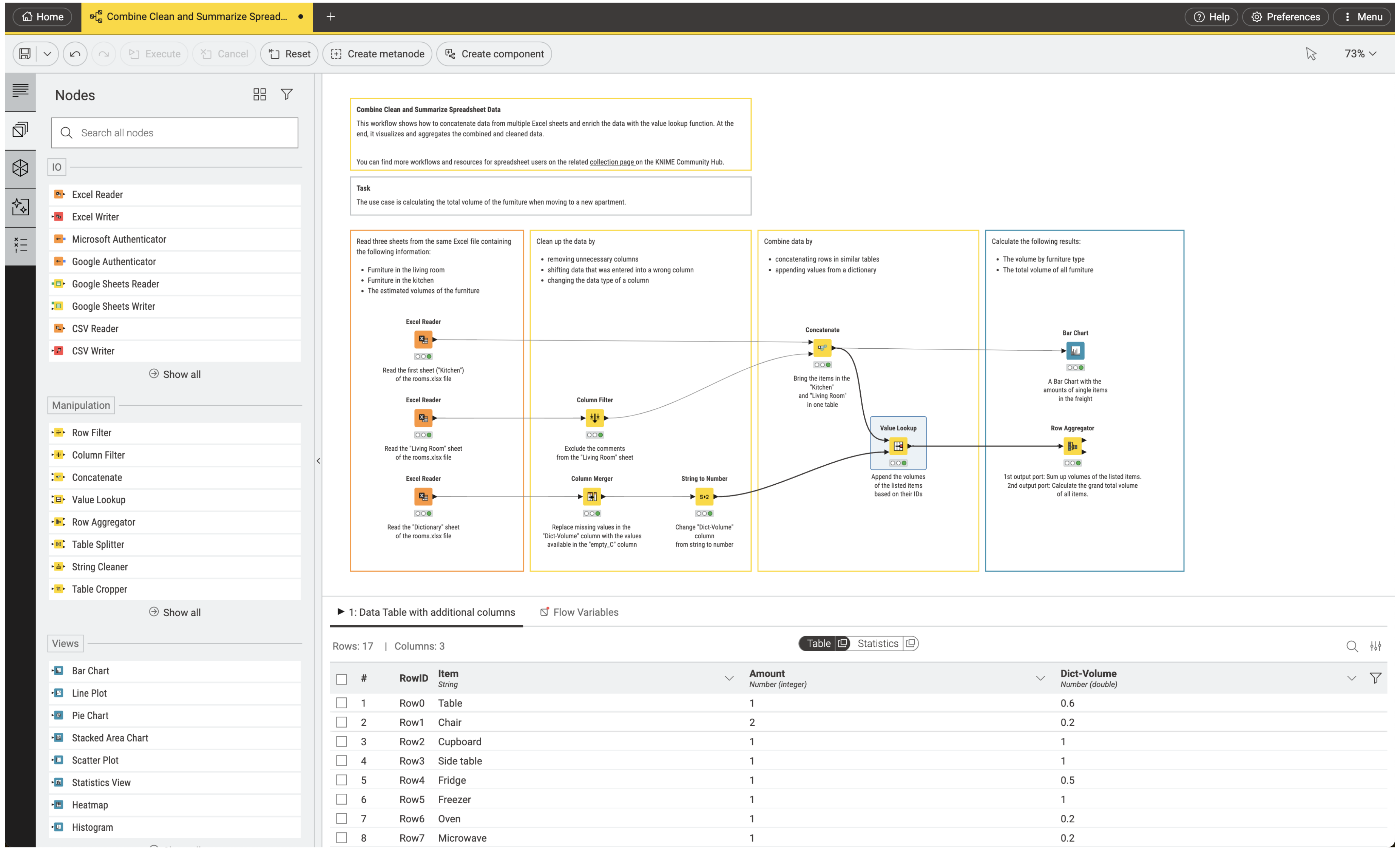
INTRODUCTION TO KNIME

Emilio Silvestri
Data Scientist, KNIME

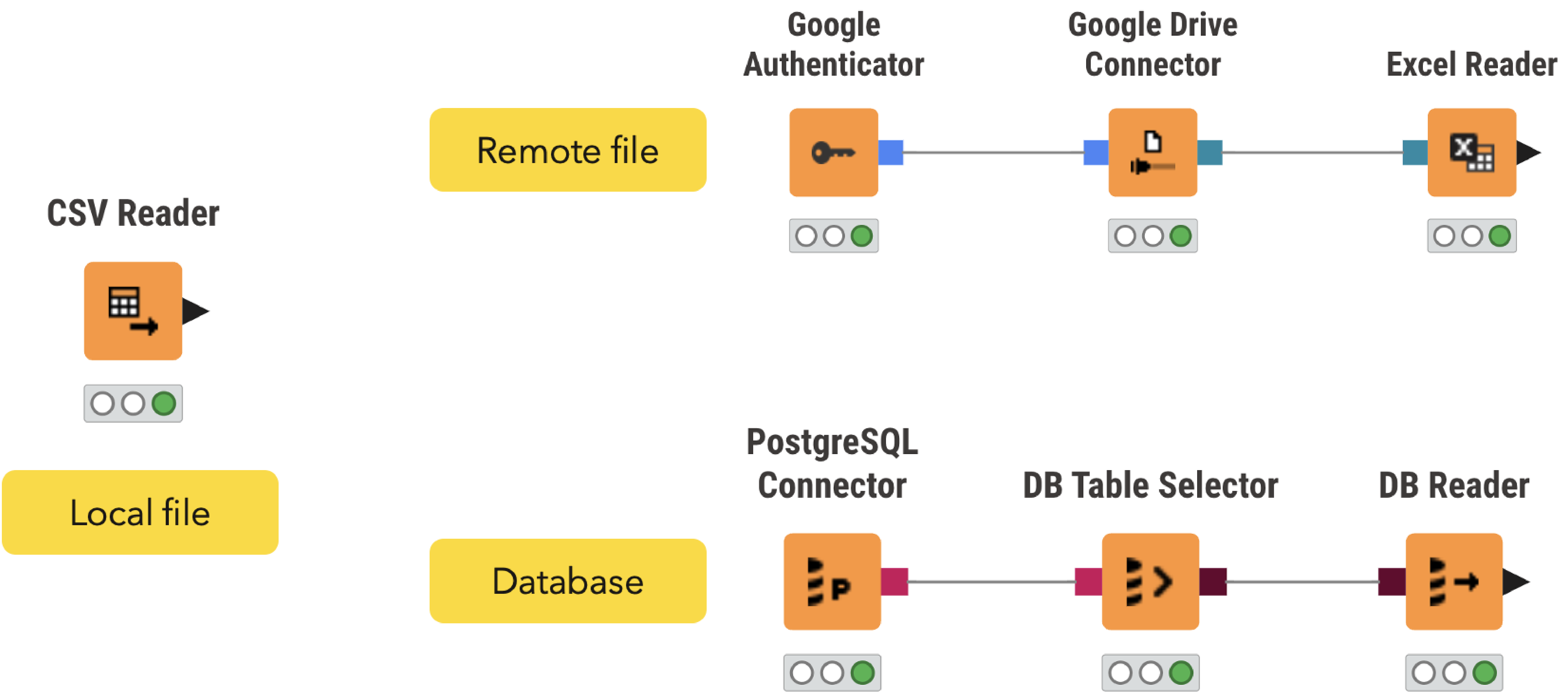
datacamp

Recap

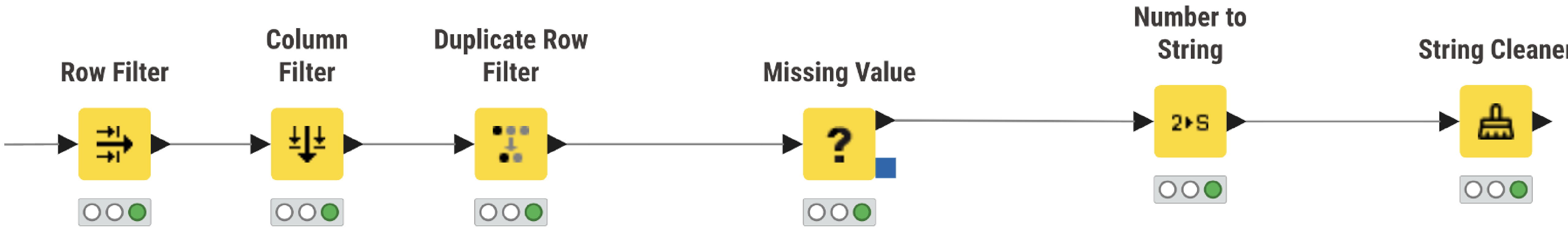
KNIME Analytics Platform



Data access


Google Authenticator
Google Drive Connector
Excel Reader
Remote file
CSV Reader
Local file
PostgreSQL Connector
DB Table Selector
DB Reader
Database

Data cleaning



Filter

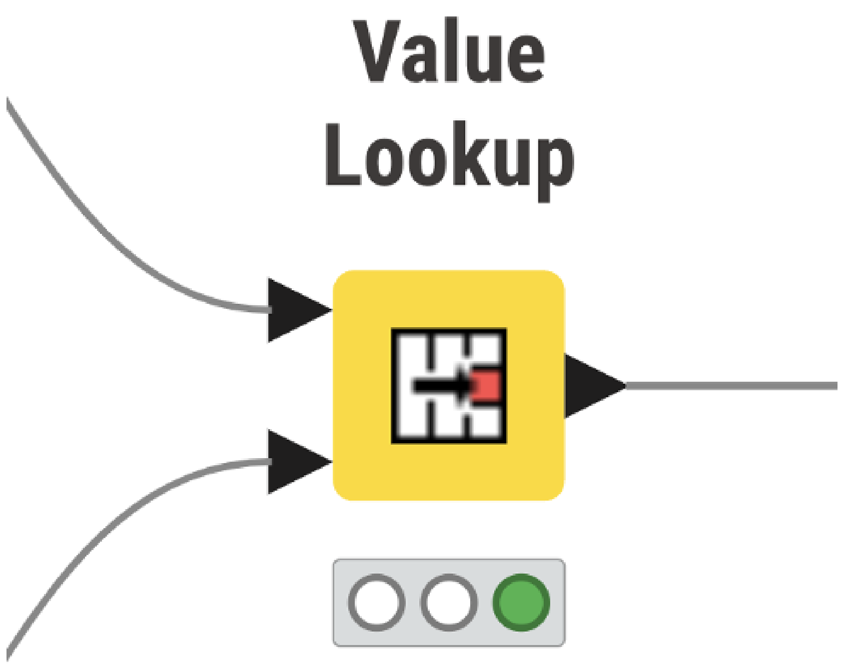
Duplicate

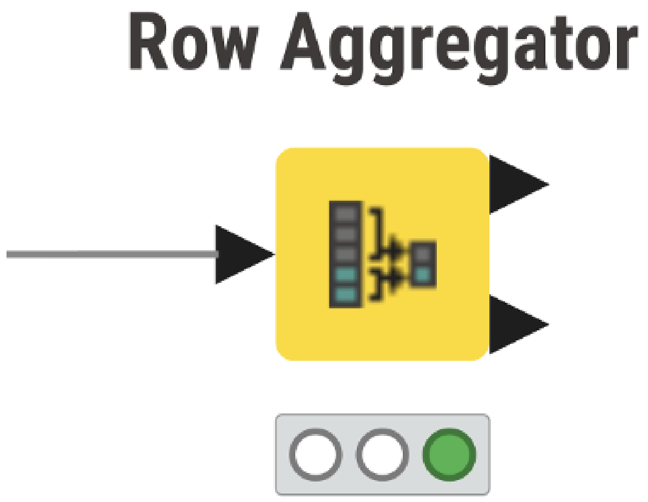
Missing values

Convert

Clean strings

Data merging and aggregation





More to come

Congratulations!

INTRODUCTION TO KNIME